

Harun Slahaldin Omar

Software Engineering Student

Laval, QC | harunager@outlook.com | (438) 508-5226 | LinkedIn | GitHub | Portfolio

EDUCATION

- **Master of Engineering, Software Engineering — Concordia University** Jan. 2027 – May 2028
Co-op Option Expected Graduation: May 2028
- **Bachelor of Engineering, Software Engineering — Concordia University** Sept. 2022 – Present
GPA: 3.3 Expected Graduation: Fall 2026
- **DEC in Pure and Applied Science — Vanier College** Sept. 2020 – May 2022

PROJECTS

- **MeMantra — Mental Wellness Mobile Application (Capstone)** github.com/MeMantraa/MeMantra
 - Co-developed a full-stack **React Native/Node.js** mental wellness app in an **11-person Agile team** across a full academic release cycle, contributing from UI components to database architecture.
 - Architected the core bottom tab navigator and integrated frontend with backend RESTful APIs; wrote unit and integration tests with **Jest** and **Supertest** to validate API endpoints.
 - Resolved critical system issues including a database migration (Oracle to PostgreSQL) and dead code elimination in JWT utilities; contributed to **CI/CD** pipelines with **Docker**, **SonarQube**, and **CodeQL** static analysis.
- **Vessel — E-Commerce Portfolio Demo** github.com/omar-haha/rcca
 - Built a full-stack e-commerce demo in **Next.js 15/React 19/TypeScript**, adapted from a production project, with server-side price re-derivation and live stock validation at checkout backed by **Supabase** (PostgreSQL).
 - Implemented a type-checked bilingual (EN/FR) UI, an order-matched “Verified Buyer” review system, and consent-gated **Google Analytics** compliant with Law 25/PIPEDA.
 - Documented a self-hosted deployment pipeline (**Docker**, **Caddy**, **GitHub Actions**) alongside the live **Vercel** deployment, demonstrating both hosting paths used in production.
- **Sillage — Event-Driven Multi-Asset Portfolio Engine** *In development*
 - Built an event-driven systematic portfolio platform in **Python/FastAPI** with a **React/TypeScript** dashboard, using one shared engine for backtesting and live execution across ETFs and crypto.
 - Prevented look-ahead bias structurally through explicit **as_of** data boundaries and next-open execution; implemented exact decimal accounting, pre-submission order journaling, idempotency IDs, position reconciliation, stale-data checks, and risk limits.
 - Developed held-out, sensitivity, timing-luck, and cost-stress validation; achieved a **0.91 Sharpe** and **19.9% max drawdown** from 2005–2026 by blending momentum with 60/40 (IBKR paper verification pending).

EXPERIENCE

- **Purolator Inc., Delivery Driver** *April 2025 – Present* Laval, QC
 - Execute time-critical delivery operations handling **200+ daily packages**, consistently maintaining a **99%+ on-time rate** across dynamic urban and suburban routes.
 - Independently resolve logistical issues (routing, access, labelling) on the fly and optimise daily load organisation to improve route efficiency.
 - Manage high-value shipments with meticulous tracking and professional client-facing service in a fast-paced, high-accountability environment.
- **Vanier College, Tutor — Calculus I, Calculus II & Web Development** *Aug. 2021 – May 2022* Montréal, QC
 - Tutored **25+ students** through weekly one-on-one and group sessions, helping raise average exam scores by approximately **13%** via personalised learning plans.
 - Developed structured review materials and targeted practice problems for Calculus and intro programming courses.

PROGRAMMING SKILLS

Languages: Python, Java, JavaScript / TypeScript, C++, C, C#, SQL

Frameworks & Libraries: React, React Native, Next.js, FastAPI, Node.js, Pandas, Plotly

Tools & Platforms: Git, Docker, Linux, PostgreSQL, SQLite, DuckDB, pytest, CI/CD, Supabase

Spoken Languages: English, French

ADDITIONAL

Interests: Piano, weightlifting, motorcycles, travel (10+ countries)